2026 WAKE FOREST SCHOOL OF MEDICINE BUILD-A-THON

Is this the same heart?

Matching patients across two Lead I ECGs, from hospital machines to Apple Watches

Team AI Deacons, Wake Forest MSBA

Brian, Nathan, D'Marley, Amy

WHY IT MATTERS

Same heart, two very different devices



Real recordings from the public St. Jude dataset. Scores are from our final model.

The goal

Doctors want to follow heart health at home with smartwatch ECGs. That only works if each watch recording can be trusted to belong to the right patient.

Our part

Our model compares a watch ECG with the patient's hospital ECG and scores how likely they came from the same person.

Also useful for

Catching ECGs filed under the wrong patient, and showing that "anonymized" ECG data can still be linked back to a person.

THE CHALLENGE

Two ECGs in, one score out

Input

Two Lead I ECGs, each up to 10 seconds at 250 readings per second. Either both from hospital machines, or one hospital ECG and one Apple Watch ECG.

Output

One score from 0 to 1 for each pair. Higher means stronger evidence that both came from the same patient. No yes/no cutoff is needed.

Judged by

AUROC on a Wake Forest holdout of patients we never see: how often a true match scores higher than a stranger pair. 1.0 is perfect, 0.5 is a coin flip.

WHAT WE BUILT

Clean it, score it, blend it

1 Prepare

Every ECG gets the same cleaning: resample to 250 Hz, filter out drift and electrical noise, cut to 10 seconds, and rescale so only the beat's shape matters, not the device.

2 Score

Two models judge each pair. The waveform model reads the full 10-second signals; the heartbeat model compares an average beat plus six measurements.

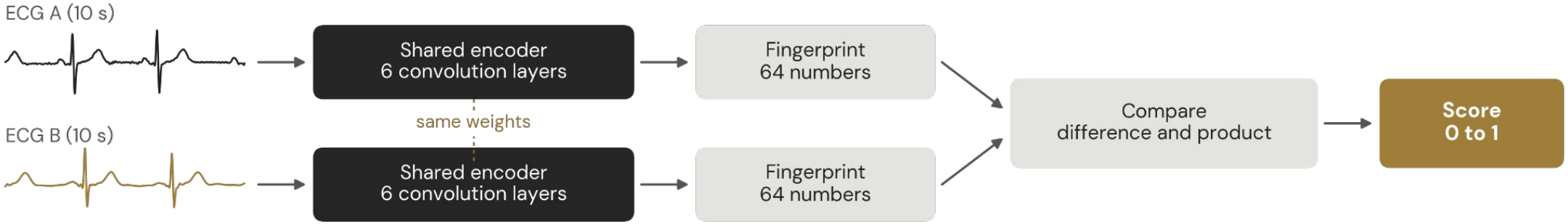
3 Blend

Final score = 80% waveform model + 20% heartbeat model. The 80/20 split was picked on a separate practice group, never on the test patients.

Our approach: A second model built on heart rate, wave heights and beat shape that a doctor can check, and Lead I checked with the formulas linking the 12 leads, not just assumed.

THE TWO MODELS

Two ways to recognize a person



Waveform model (Siamese CNN), 80%

Both 10-second ECGs run through the same encoder with shared weights (6 convolution layers), which turns each into a 64-number fingerprint. A small layer compares the two fingerprints and returns a score. We train three copies from different random starts and average them.

Heartbeat model (feature net), 20%

Finds every beat, lines them up and takes the median beat, so noise on one or two beats cancels out. Adds six measurements: heart rate, heart rate variability, R peak height, S wave depth, T wave height and beat consistency. A small network (16 neurons) compares the two summaries.

One model finds its own patterns; the other uses measurements a doctor would recognize, so a doctor can check why two ECGs were matched.

HOW WE TRAINED IT

Learn from pairs, test on strangers

37,237

hospital ECGs from 10,000 MIMIC-IV patients, Lead I only. Every patient has 2+ ECGs (up to 5 used), so true matches exist.

243

St. Jude patients with a same-day hospital ECG and Apple Watch ECG: 195 to train, 48 locked for the test.

3 to 1

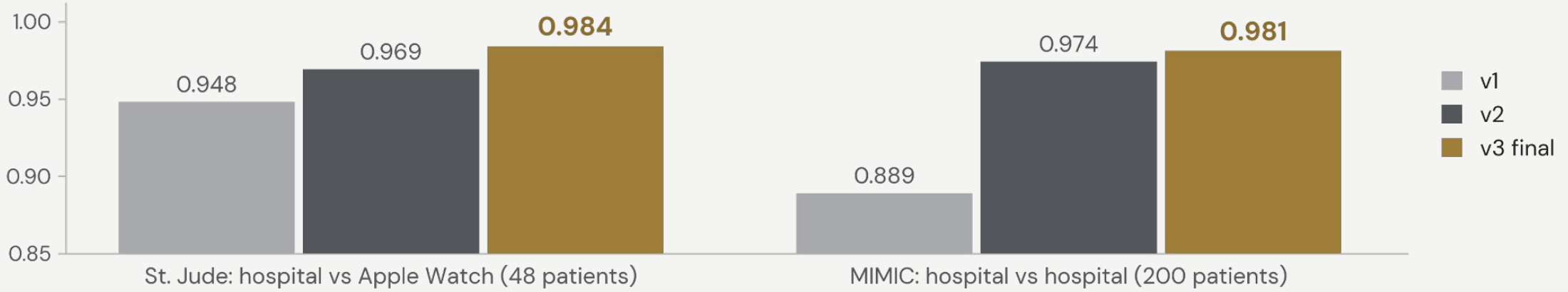
stranger pairs for every true-match pair, with new pairs drawn in each of the 100 training rounds.

Patients were split before any pairs were built, so no one appears in both training and testing. Locked for the final test: 48 St. Jude and 200 MIMIC patients. Every setting was chosen on a separate practice group of 39 St. Jude and 300 MIMIC patients.

HOW WELL IT WORKS

About 0.98 on patients the model never saw

Locked-test AUROC (1.0 = perfect, 0.5 = coin flip). St. Jude 95% range: 0.966 to 0.995.



79.2%

Top 1, St. Jude: the patient's own watch ECG scored highest out of 48 (chance ~2%)

68.5%

Top 1, MIMIC: the patient's own second ECG scored highest out of 200 (chance 0.5%)

Top 1: we compare one patient's ECG with every ECG in the test group. If the highest score goes to that patient's own ECG, the true match, it counts as correct.

WHERE IT FAILS

Two different reasons for misses

Hospital vs watch

92%

of St. Jude ranking mistakes came from noisy watch recordings. The single worst miss combined a messy tracing with a 32 bpm heart rate jump between the two recordings.

Hospital vs hospital

81% to 35%

MIMIC top 1 accuracy when the two ECGs' heart rates are within 5 bpm, versus when they differ by 20 bpm or more. A big heart rate change makes the same person look different.

WHAT A HARD WATCH CASE LOOKS LIKE

Clean Apple Watch recording



Noisy Apple Watch recording



Both findings come from our error analysis on the practice group. Example traces: two real Apple Watch recordings from the public St. Jude data.

SANITY CHECKS

We checked whether it was too good to be true

Learning helps

A simple rule that compares typical (median) beat shapes, with no learning, scores 0.803 on the same 48 locked patients. Our model scores 0.984.

Right lead

The St. Jude files and the dataset paper never say which channel is Lead I. But the 12 leads are connected by fixed formulas, so we used them as a test. The formulas only add up if the first channel is Lead I, and they did for all 243 patients.

Reproducible

We reran the submitted model files on a fresh install: the same 0.984 AUROC and 79.2% top 1. Demo pair: same person 1.000, stranger 0.050.

The baseline and reproducibility checks use the 48 locked patients, never used for training or tuning. The lead check covers all 243 St. Jude files.

WHAT WE TRIED THAT DID NOT HELP

We tested fixes before trusting them

Idea	What it was meant to do	Result (never on the locked test)
Quality-aware blend	Lean more on the waveform model when beats look inconsistent, and drop odd beats first	No gain: nothing beat the 80/20 blend's 0.972 (St. Jude) and 0.985 (MIMIC); top 1 went down
Heart rate stretching	Stretch or squeeze ECGs by up to 30% so a faster or slower heart isn't read as a new person	Slightly worse: -0.004 (St. Jude), -0.001 (MIMIC); lower top 1 in every heart rate group
Gentler or heavier cleaning	Check that filtering and scaling weren't erasing identity	All within 0.01 AUROC, so we kept the standard setting
Other parts of the watch ECG	First, last, or an average of three 10 s windows instead of the middle	Middle 10 s did as well or better: 0.972 vs 0.967 to 0.972

We kept v3 because nothing beat it. None of these tests used the locked test patients.

HONEST LIMITS

What our numbers do not prove

Small, reused tests

Only 48 watch patients, and the same 48 scored every version. The 95% range for 0.984 is 0.966 to 0.995, so read it as about 0.98.

Same-visit watch pairs

All St. Jude hospital and watch pairs were recorded the same day, 98% within two hours. Pairs taken days or months apart may score lower.

Ranking is easier than picking

AUROC asks whether a true match outscores a stranger. Real use means picking one person out of many: our top 1 was 79.2% among 48 and 68.5% among 200.

Untested unknowns

A new hospital and population, other watch models, changes from illness, and differences across age, sex and race.

CAN THIS WORK IN THE FIELD?

A flagging aid, not an identifier

Plausible, with review

- Flag ECGs filed under the wrong patient
- Audit a dataset for re-identification risk before sharing
- Link records from watch and hospital in consented research

Not ready for

- Confirming who someone is on its own
- Automatic decisions with no person checking
- Security or forensic identification

Needed first

- Testing on Wake Forest data at larger scale
- A score cutoff set so false matches stay rare
- Watch ECGs from different days, and checks across patient groups
- Privacy and ethics review

TAKEAWAYS

AUROC 0.984 hospital vs Apple Watch, 0.981 hospital vs hospital, on patients the model never saw

A single-lead watch ECG carries a personal signature that survives the switch from hospital machine to wrist: the trust check home heart monitoring needs.

More hospital data and averaging three models did the work. Careful testing kept us from shipping changes that did not help.

Thank you. Questions?

Team AI Deacons, Wake Forest MSBA